

UBC Renewable Energy Mapping: Executive Summary

Purpose

To meet British Columbia's forecasted rise in electricity demand over the next two decades and fulfill the net-zero climate commitments by 2050, the province must increase renewable energy infrastructure development. In response to this call to action, a GIS-based Analytic Hierarchy Process (AHP) analysis was performed to:

1. Identify locations across BC with the best potential for photovoltaic (PV) energy electricity generation, and
2. Determine the total capacity and yearly energy generation potential possible from the suitable sites identified.

The model includes environmental, economic, and climatic criteria, together capturing core influences on renewable energy performance, accessibility, and development costs. As a screening tool, the results can guide organizations in prioritizing areas where development is most likely to yield high energy output.

Methods

The provincial solar site suitability map was generated by reclassifying six core criteria layers (Table 1) on a scale from 5 (most suitable) to 1 (least suitable). To combine these heterogeneous factors into a composite suitability surface, weights representing the relative importance of each criterion in the siting process were derived using the Analytical Hierarchy Process (AHP). To ensure model robustness, nine domain experts spanning academia, policy, and industry contributed to the weighting process through the individual creation of pairwise comparison matrices (PCM). Aggregating these PCMs via the geometric mean method grounded the weights in real-world planning priorities. This framework effectively quantifies subjective judgements to deliver a robust multi-criteria assessment of regional solar suitability.

Table 1. The input criteria used to assess solar PV suitability across BC.

Criteria	Sub-Criteria	AHP-Derived Weight (%)
Climatic	Global Tilted Irradiance	26.3
Economic	Proximity to Substations	30.2
	Proximity to Demand Centers	14.4
	Proximity to Major Roads	11.2
Topographic	Ground Aspect	9.3
	Ground Slope	8.5

With the most optimal sites for PV development identified, the technical potential was determined for the top two suitability classes by calculating installable capacity (GW) and annual energy yield (TWh/year) for each viable site, using five parameters:

Parameter	Impact	Value
Site area (m ²)	Available space for arrays	Site dependent
Panel efficiency (η)	A measure how well a solar panel turns sunlight into usable electricity	0.23 (23%)
Latitude dependent Ground Cover Ratios (GCR)	Accounts for panel spacing and losses due to inter-row self-shading	Northern latitudes – 0.227 Central latitudes – 0.263 South latitudes – 0.314
Packing Density Ratio (R_{dt})	The ratio between direct and total land use in a designated site	0.724 (72%)
PV Power Potential (kWh/kW _p /year)	Long term yearly averages for standard crystalline silicon modules (c-Si) at a fixed optimum tilt (OPTA)	Site dependent

Technical yield results were generated for small scale solar farms (1 MW minimum production threshold) and large scale (100 MW minimum production threshold) by filtering out sites smaller than 5 acres and 500 acres, respectively. This follows the assumption that 5 acres of land is needed to generate 1 MW of solar energy.

Results

It was found that 75.5% of BC (716,261.8 km²) is restricted from development due to unsuitable land cover types (e.g. water, built environments, wetlands), prime agricultural land, dense/mature forest stands, and protected and conserved areas. The remaining 24.5% of the province was classified as followed:

Table 2. Summary of the suitability distribution across the province

Suitability Class	Area (km ²)	Portion of Total Land Base (%)
1 – Least Suitable	80,926	8.5
2 – Less Suitable	108,095	11.4
3 – Moderately Suitable	39,216	4.1
4 – More Suitable	3602	0.38
5 – Most Suitable	141	0.02

Resource yield estimates were summed across only the top two suitability classes (Class 4 and 5) as these represent the area with the most promising return on investment for solar PV development. Yield estimates (Table 3) represent the technical ceiling of energy generation and reflect a scenario where *all* sites deemed more and most suitable have installed solar arrays.

Table 3. Resource yield estimates across the top two suitability classes under two different minimum production threshold scenarios.

Scenario Class	1 MW		100 MW	
	4	5	4	5
Number of sites	20,029	993	119	1
Area of sites (km ²)	2750.4	93.9	545.2	2.2
Capacity (GW)	138.1	4.7	27.7	0.1
Total Energy (TWh)	176.3	6.1	36.3	0.2

Dissecting the results further, it was found that the Thompson-Nicola, East Kootenay, and Peace River regional districts hold the greatest amount of solar PV potential, as they achieved the biggest sum of Class 4 and 5 land, installable capacity, and annual energy generation out of the province.

Table 4. Summary of results for the top-ranking regional districts in terms of both geographical and technical potential. Results correspond to a 1MW production threshold.

Rank	Regional District	Area of Viable Sites (km ²)	Capacity (GW)	Annual Energy (TWh)
1	Thompson-Nicola	708.7	37.1	48.8
2	East Kootenay	460.8	24.1	32.0
3	Peace River	245.7	10.2	13.3

Conclusions

This project makes four key contributions to solar energy planning; first, it uses an expert-informed GIS-AHP framework, incorporating extensive engagement feedback from academics, industry professionals, and government/policy personnel into model parameters. Second, it advances beyond traditional suitability mapping by translating high-suitability areas into quantified annual energy generation estimates. Third, it offers an open-source decision-support tool which can be modified to fit future research needs (accessible through [github](#)). Lastly, it is the first solar PV suitability analysis for British Columbia - providing a tool to support the provinces transition to a diversified renewable energy portfolio.

Future Research

Future work should bridge spatial land-use suitability with electrical infrastructure constraints by considering utilization rate information. This would allow the model to prioritize substations that have greater availability than those that are at peak capacity. Another beneficial expansion would investigate and implement the effects of seasonal curtailment and battery storage options into the suitability model to further refine generation estimates. Ultimately, combining this GIS-based, land-use dominant evaluation, with dynamic grid modelling will transition the tool from regional site screen to an operation asset for energy developers.

